

EDUCATION

Lund University, Sweden Exchange Program; Master’s in Computer Science Erasmus+ Scholar; About €6000 in grant.	Lund, Sweden Jan – Jun 2026
University of Ghana, Ghana MPhil Computer Science; CGPA 3.75/4.0 Thesis: A Residual Hybrid Deep Learning Framework for Missing Data Imputation in IoT Sensor Data Using LSTM and Variational Autoencoders	Accra, Ghana Oct 2024 – Oct 2026
University of Ghana, Ghana Bachelor’s in Computer Science; CGPA 3.45/4.0 Thesis/Project: SmartFlow: IoT-Based Smart Irrigation and Environmental Monitoring System	Accra, Ghana Oct 2019 – Aug 2023

SKILLS SUMMARY

• Languages:	Python, SQL, JavaScript, Dart, HTML
• Frameworks:	Django, FastAPI, Flutter, PyTorch, TensorFlow, ScikitLearn
• Tools:	Git, Postman, PostgreSQL, PgAdmin, LLM, Agentic AI, Docker, REST API
• Platforms:	GitHub, Jupyter Notebook, VSCode, Codex, DigitalOcean, Linux Servers
• Soft Skills:	Rapport Building, Team Playing, People Management, Excellent Communication

RESEARCH INTERESTS

- AI-Enabled Software Engineering (AI4SE, SE4AI, etc.)
- Software Security and Secure Software Development
- Empirical Software Engineering/Development, Testing, and Optimization
- Artificial Intelligence (AI) and Machine Learning Systems Engineering (MLOps)
- Theoretical Computer Science

RESEARCH EXPERIENCE AND PUBLICATIONS

Research Experience	
• Graduate Teaching and Research Assistant (TA/RA), University of Ghana	2023 – Present
• MPhil Thesis Researcher, University of Ghana	2024 – 2026
(1) Book Chapter: Kyeremanteng, P. S., et al. (2025) Federated Learning at the Edge: Enhancing Privacy and Efficiency in Distributed AI Systems Publisher: IntechOpen DOI: 10.5772/intechopen.1011986	
(2) Manuscript: Kyeremanteng, P. S., et al. PoAD-Lite: A Lightweight, Activity-Aware Consensus Protocol for Scalable and Energy-Efficient IoT Blockchains Publisher: IET Smart Cities – IEEE Manuscript ID: SMC-2025-10-0088.R2 Under Review	
(3) Manuscript: Kyeremanteng, P. S., et al. Secure Sharding in PoAD: Enhancing Scalability Without Security Trade-offs Publisher: IET Blockchain – IEEE Manuscript ID: BLC-2025-08-0170 Under Review	

CONFERENCES AND PRESENTATIONS

(1) Postgraduate Research Workshop, Computer Science Department – University of Ghana (2025)

Presentation: Missing Data Imputation Techniques: Applications, Challenges, and Innovations

Award: 3rd Best Paper Award

(2) Nyansapo AI Symposium on Sustainable Development

Organized by the Ghana Atomic Energy Commission and Université Paris-Saclay (2023)

Presentation: AI-Based predictive system for electricity outage forecasting.

WORK EXPERIENCE

Teaching and Research Assistant | [University of Ghana](#)

October 2023 – Present

- Conducted tutorials on machine learning, heuristic search, and knowledge representation.
- Assisted in research in the labs of two professors during my time as a research assistant.
- Assisted in teaching Artificial Intelligence, database management, and system design.

Lead Software Engineer | [AfriLearnTech Ghana Limited](#)

September 2025 – Present

- Led the development of a national digital learning platform for Liberia's Ministry of Education.
- System includes LLM Agents for course content generation to augment localized content.
- System includes (synchronizable) offline boxes to be deployed in schools with low connectivity.

Founder | PKay Software Consultancy

October 2023 – Present

- Led the development of software systems, including web and mobile, for a wide range of clients.
- Systems include Admission Systems, Accountability (Buddy) Systems, and Business Portals.
- Handled stakeholder engagements and the entire Software Development Life Cycle.

Software Developer | PayHub Ghana Limited

April 2021 - December 2025

- Developed Payment Processing APIs for internal use and for clients.
 - Developed fintech-grade dashboards for transactions monitoring and management.
 - Reduced system latency through Redis-based caching and optimized session management.
-

PROJECTS

Projection Software | [Link](#)

April 2026

- Built a desktop church projection platform using Tauri, React, and SQLite.
- Developed a real-time dual-window projection and display engine.
- Engineered offline content management for scriptures, hymns, and announcements.

AfriLearn Digital Learning Platform | [Link](#)

Oct 2025 - April 2026

- Built an offline-first national digital learning platform for Liberia's Ministry of Education.
- System includes easy-to-deploy hardware 'boxes' deployable in low connectivity areas.
- Automatic updation and synchronization of educational content (videos, Audio, PDFs, etc.).
- Uses LLM to suggest and generate personalized lessons and assessments for learners.

Synther | [Link](#)

April 2026

- A synthetic data generation tool for generating highly configurable datasets
- Implemented Celery workers and Redis for scalable background generation and live progress.
- Designed an intuitive wizard with datatype-aware fields, options, and downloads.
- Ensured data fidelity with numpy/Faker-based generators, and deterministic seeding.

Nrf24 Firewall FullStack and Embedded | [Link](#)**March – April 2026**

- Designed/implemented a lightweight software-defined firewall for an NRF24-based IP network.
- Built a full-stack system using FastAPI and React with a staged rule engine and web interface.
- Configured Linux TUN interfaces, routing, NAT, and integrated nftables for traffic filtering.
- Evaluated system performance using UDP throughput and latency measurements with iperf3.

ReviewMate | [Link](#)**November 2025**

- Designed and built a Word add-in with an LLM-powered review agent (TypeScript, Office.js).
- Designed prompt-driven review workflows to generate structured, actionable feedback as review comments in Word.
- Integrated OpenAI-compatible LLM pipeline with JSON outputs and resilient parsing.
- Added AI writing tools (trim, paraphrase, summaries) with secure API key handling.

Admissions Management System | [Link](#)**January 2024 – Present**

- Designed/developed an admissions management system for [GIMPA](#) (a tertiary institute in Ghana)
 - The system includes a management portal for monitoring and management, with an automated and manual notification system.
 - Over 2000 applications processed so far, and counting.
 - Application processing time is improved by over 500 percent.
-

REFEREES**(1) Prof. Yaokumah Winfred**

Computer Science Department, University of Ghana
wyakumah@ug.edu.gh || +233 24 428 3488

(2) Prof. Charles Teye Amoatey

Director, Africa Center for Development Impact (ACDI)
Professor, Ghana Institute of Management and Public Administration
camoatey@gimpa.edu.gh || +233 24 975 9944

(3) Prof. Kofi Sarpong Adu-Manu

Computer Science Department, University of Ghana
ksadu-manu@ug.edu.gh || +233 24 460 2374

(4) Dr. Gyamfi Nana Adwabour

Glocal CEO, AfriLearnTech Company Limited
dr.gadwabour@afrilearntech.com || +233 54 030 7039